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The gait abnormality index: A summary metric for three-dimensional gait analysis

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ABSTRACT

Background: This paper proposes an easy to calculate and adaptable summary gait metric, the Gait Abnormality Index (GAI), which is capable of simultaneously including kinematic and kinetic data, overcoming a key limitation of existing metrics.

Research question: To determine the validity, reliability and sensitivity of the GAI.

Methods: The GAI is calculated by averaging Gait Abnormality Scores, which are normalised distance metrics used to describe the deviation of pathological gait data from that of healthy controls. Validity was assessed using Pearson's correlation analysis to explore relationships between the GAI and the Gait Profile Score. Test-retest reliability of the GAI was assessed using intra-class correlation coefficients (ICC) and standard error of the measurement (SEM), and data from total hip arthroplasty patients. An independent samples t-test was used to compare GAI scores between knee osteoarthritis and total hip arthroplasty patients to explore the metrics sensitivity.

Results: A strong positive correlation ($r \geq 0.896$; $p < 001$) was reported between the GAI and the Gait Profile Score. Good test-retest reliability (ICC =0.830) was reported for the GAI. Knee osteoarthritis patients displayed significantly ($p = .017$; Hedge's g effect size = 0.98) greater GAI scores compared to total hip arthroplasty patients, with the mean difference (0.34 a.u) above the SEM (0.15 a.u).

Significance: The GAI offers an easy to calculate summary metric for three-dimensional gait analysis, which displays good validity and reliability, and is sensitive to different pathological conditions.

1. Introduction

Three-dimensional gait analysis (3DGA) is commonly used to understand the influence of neuromuscular or musculoskeletal conditions and therapeutic, surgical or training interventions upon lower limb joint function. 3DGA provides large quantities of objective data regarding joint function and insight into the muscular control of the movement, if relevant kinetic parameters such as moments and powers are calculated. However, the complexity and quantity of data generated has been highlighted as a factor limiting the application of 3DGA within clinical settings [1]. As such summary metrics have been developed, providing a single value to describe the 'quality' of movement or a global representation of the difference between pathological and normal asymptomatic joint motion or loading patterns [1–4].

A number of summary gait metrics for 3DGA have been developed and are reviewed in-depth by Cimolini and Galli [1], and Gonçalves, Lama and da Silva [5]. Traditional summary gait metrics were developed using either kinematic [2,3] or kinetic [6] data only. More contemporary metrics have been developed which are derived from kinematic and kinetic gait data [7–9]. However, all of these metrics utilise varying multivariate statistical procedures, which increases the complexity of adapting these beyond the populations, biomechanical models, and movement tasks with which they were originally developed. To overcome this limitation the Gait Profile Score (GPS) was developed by Baker et al., [4]. Specifically, the GPS is derived from a series of Gait Variable Scores (GVS), which represent the root mean square difference between pathological and healthy movement patterns, respectively. Furthermore, the GVS develop a Movement Analysis Profile (MAP), providing a means of identifying the extent to which different kinematic outputs contribute to the overall GPS [4]. However, like other traditional summary metrics [2,3], the GPS contains only kinematic data and the ability to incorporate kinetic data is limited by the use of root mean square difference due to differing units of measure (degrees and Nm/kg).

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This paper proposes a means of advancing the GVS, GPS and MAP concepts to enable the simultaneous inclusion of kinematic and kinetic data within a novel and adaptable summary metric for 3DGA. Specifically, the Gait Abnormality Index (GAI) is proposed, which utilises z-scores to provide an easy to calculate and interpret summary gait metric. Additionally, the MAP is expanded to provide a visual means of exploring the GAI over the entire walking gait cycle, enabling the timing of larger deviations from normal to be identified. The aim of this study was to explore the validity, test-retest reliability and sensitivity to different pathological conditions of the GAI.

2. Methods

2.1. Participants

Retrospective analysis of existing data available to the researchers was undertaken within this study. Specifically, 20 healthy participants, defined as individuals with no known neurological or musculoskeletal conditions at the time of testing, acted as a control group. Data from 20 knee osteoarthritis patients who were awaiting total knee arthroplasty, and 10 individuals who had undergone total hip arthroplasty at least 12 months prior to data collection formed pathological groups for the development and exploration of the GAI. Descriptive characteristics of each participant group are detailed in Table 1. All participants provided written informed consent, which included consent for secondary data analysis, prior to undertaking data collection. Ethical approval for the original studies in which the data were collected was provided by the Health Research Authority (17/LO/1584 & 20/NS/0057).

2.2. Procedures

Healthy control and knee osteoarthritis groups completed a single testing session, while the total hip arthroplasty group completed test and retest sessions at least 1 week apart. All participants completed overground walking trials, barefoot along a 7 m walkway, at a self-selected velocity (Table 1). A 10-camera motion capture system (Qualisys, Gothenburg, Sweden) and two synchronised force platforms (Kistler, Winterthur, Switzerland) operating at 200 Hz and 2000 Hz respectively, captured kinematic and kinetic data. A six-degrees-of-freedom model was used to track the position of the pelvis, thighs, shanks, and feet of participants to enable hip, knee and ankle joint kinematics and kinetics to be calculated. A detailed description of the six-degrees-of-freedom model and the reliability of associated joint kinematics can be found in Langley et al., [10].

2.3. Data processing

Marker trajectories were reconstructed, gap filled to a maximum of 10 frames, and labelled within Qualisys Track Manager (version 2018.1, Qualisys, Gothenburg, Sweden). Trials with gaps wider than 10 frames were discarded during data processing. Five processed trials for each participant were exported as C3D files to Visual 3D (C-Motion Inc, Germantown, MD, USA). Kinematic and kinetic data were low pass filtered at 6 Hz and 24 Hz, respectively. Joint angles and moments at the hip, knee and the ankle, in all three planes, were calculated using an XYZ cardan sequence of rotations and Newton-Euler equations. Initial contact and toe off were identified using a 10 N threshold applied to the vertical ground reaction force data, and the subsequent initial contact using the automated gait events within Visual 3D. Joint angles and moments were time normalised to 100% gait cycle duration.

Table 1
Participant characteristics for the healthy control, knee osteoarthritis and total hip arthroplasty groups. Mean (SD).

	Healthy Control	Knee Osteoarthritis	Total Hip Arthroplasty
n	20	20	10
Age (years)	58 (9)	68 (5)*	71 (7)*
Height (m)	1.66 (0.08)	1.68 (0.08)	1.66 (0.13)
Mass (kg)	73 (16)	88 (15)*	80 (20)
Gender (M:F)	10:10	13:7	7:3
Walking Velocity (m/s)	1.36 (0.13)	0.92 (0.19)*	1.12 (0.33)*†
Gait Cycle Duration (s)	1.02 (0.09)	1.35 (0.16)*	1.11 (0.15)†
Toe Off (%)	60 (1)	63 (3)*	60 (2)†

* Significantly ($p < .05$) different to healthy control group
† Significantly ($p < .05$) different to knee osteoarthritis group

2.4. Gait abnormality index calculation

The GAI calculation is similar to that of the GPS [4]. Initially, Gait Abnormality Scores (GAS) are calculated using Eqs. 1 and 2, for each participant, at each joint, in each plane, at each percentage of the gait cycle.

$$d_i = \sqrt{(\overline{H_i} - p_i)^2} \tag{1}$$

$$GAS_i = \left|\frac{\overline{dH_i} - d_i}{\sigma H_i}\right| \tag{2}$$

Where; d = Euclidean distance, i = percentage of gait cycle, $\overline{H}$ = healthy control group mean, p = relevant participants data, σH = standard deviation of the healthy control group data.

To reduce the sensitivity of the GAS when σH was near zero, for instance during the swing phase where ankle joint moments for all participants are near zero, which increases the magnitude of GAS_i as calculated using Eq. 2, an if function was built into the calculation. The function states that *if* $\sigma H_i < 0.01$, then $GAS_i = d_i$. A modified version of the Movement Analysis Profile (MAP) can then be calculated by either plotting the GAS over the walking gait cycle, or by calculating the average and standard deviation of the GAS over the gait cycle. Finally, the GAI is calculated by averaging the GAS. Kinematic only, kinetic only and combined versions of the GAI are presented within this study.

2.5. Data analysis

All analysis was undertaken within SPSS Statistics (Version 28.01.0, IBM Corporation, Armonk, New York, USA). The alpha level used to identify statistical significance was set to $p < .05$. Data analysis was undertaken in three parts; initially, the knee osteoarthritis group data was used to explore the concurrent validity of the GAI, by quantifying the relationship between the kinematic only and combined GAI metrics with the GPS, using Pearson's correlation coefficients (r). The rationale for including both the kinematic only and combined GAI metrics was due to the GPS being calculated using kinematic data alone. Furthermore, the relationship between the combined GAI and walking velocity was explored, using Spearman's correlation coefficients (*rho*), as data violated parametric assumptions.

Secondly, the test-retest reliability of the GAI was quantified using the total hip arthroplasty patient's data. Specifically, intra-class correlation coefficients (ICC) for absolute agreement and the standard error of the measurement (SEM) were calculated from the output of a two-way mixed effects model analysis of variance (ANOVA) [11,12]. ICC and SEM values were calculated to provide both relative and absolute measures of reliability. ICC values were interpreted as follows: < 0.5 = poor, $0.5 - 0.75$ = moderate, $0.75 - 0.9$ = good, and > 0.9 = excellent reliability [13].

Finally, the sensitivity of the GAI was assessed by comparing GAI scores between the knee osteoarthritis and total hip arthroplasty groups using an independent samples t test. Effect sizes were estimated for this comparison using Hedge's g, and interpreted as follows; small < 0.5,

medium ≥ 0.5 and < 0.8, large ≥ 0.8 [14]. For this comparison data from the test (first) assessment of the total hip arthroplasty group was utilised.

3. Results

Fig. 1 displays the GAS and the resulting GAI for one knee osteoarthritis patient. Fig. 2 displays the modified MAP showing the average GAS and GAI over the walking gait cycle for the same knee osteoarthritis patient. Hip, knee and ankle joint kinematic and kinetic profiles over the walking gait cycle for the knee osteoarthritis patient and healthy control group are displayed in Supplementary Figures 1 and 2.

3.1. Correlations with GPS and walking velocity

Statistically significant strong positive correlations ($r \geq 0.896$; $p < .001$) were reported between both the kinematic and combined variations of the GAI and the GPS (Fig. 3A & B). A weak non-significant negative correlation ($rho = -0.171$; $p = .470$) was reported between the GAI and walking velocity (Fig. 3C).

3.2. Reliability

The GAI displayed good reliability (ICC =0.830; 95% CI =0.326 –0.958) across test (mean ± SD: 1.18 ± 0.27 a.u.) and retest (1.23 ± 0.31 a.u.) sessions. The SEM for the GAI was 0.15 a.u. Supplementary Figure 3 displays the GAS and resulting GAI over the walking gait cycle for the test and retest sessions. Hip, knee and ankle joint kinematic and kinetic profiles over the walking gait cycle during the test and retest sessions for the total hip arthroplasty patient are displayed in Supplementary Figures 4 and 5.

3.3. Sensitivity

GAI values were above 1 for both the knee osteoarthritis and total hip arthroplasty patients, demonstrating abnormal gait mechanics in both populations (Fig. 4). GAI was significantly ($p = .017$) larger for the knee osteoarthritis patients compared to the total hip arthroplasty patients, with the mean difference of 0.34 above the SEM and associated with a large effect size ($g = 0.98$). Supplementary Figure 6 displays the GAS and resulting GAI over the walking gait cycle for the knee osteoarthritis and total hip arthroplasty groups. Hip, knee and ankle joint kinematic and kinetic profiles over the walking gait cycle for each group are displayed in Supplementary Figures 7 and 8.

4. Discussion

This study proposes a novel summary gait metric, the GAI. The GAI overcomes one of the key limitations of previous summary gait measures by enabling the simultaneous inclusion of both kinematic and kinetic data within an easy to calculate metric. Furthermore, by maintaining the MAP concept [4] the contribution of individual joints, in each plane, to the overall GAI can be explored. Plotting the GAS over the walking gait cycle to expand the MAP also highlights the timing of larger abnormalities in joint angle or moment time-histories, increasing the temporal resolution of the data. The findings demonstrate that the GAI is reliable and sensitive to different pathological conditions, while also demonstrating high levels of concurrent validity.

The GAI is based upon the same fundamental quantity as the Gait Deviation Index (GDI) [3] and GPS [4], i.e., the Euclidean distance between pathological gait data and healthy controls. The differences between the three metrics primarily lies in the initial feature selection and/or standardisation procedures. The GDI is the only metric to have a specific feature selection process, with principal component analysis used to identify 15 gait features which can be used to describe the majority of the variance in the original data set. This initial feature selection process is one of the factors limiting the easy application of the GDI within clinical practice [1,4]. The GPS removed the feature selection and standardisation procedures inherent within the GDI calculation, thus providing a more simplistic metric for use within applied settings. However, reporting the data in its raw form as the root mean square difference measured in degrees, results in an inability to simultaneously

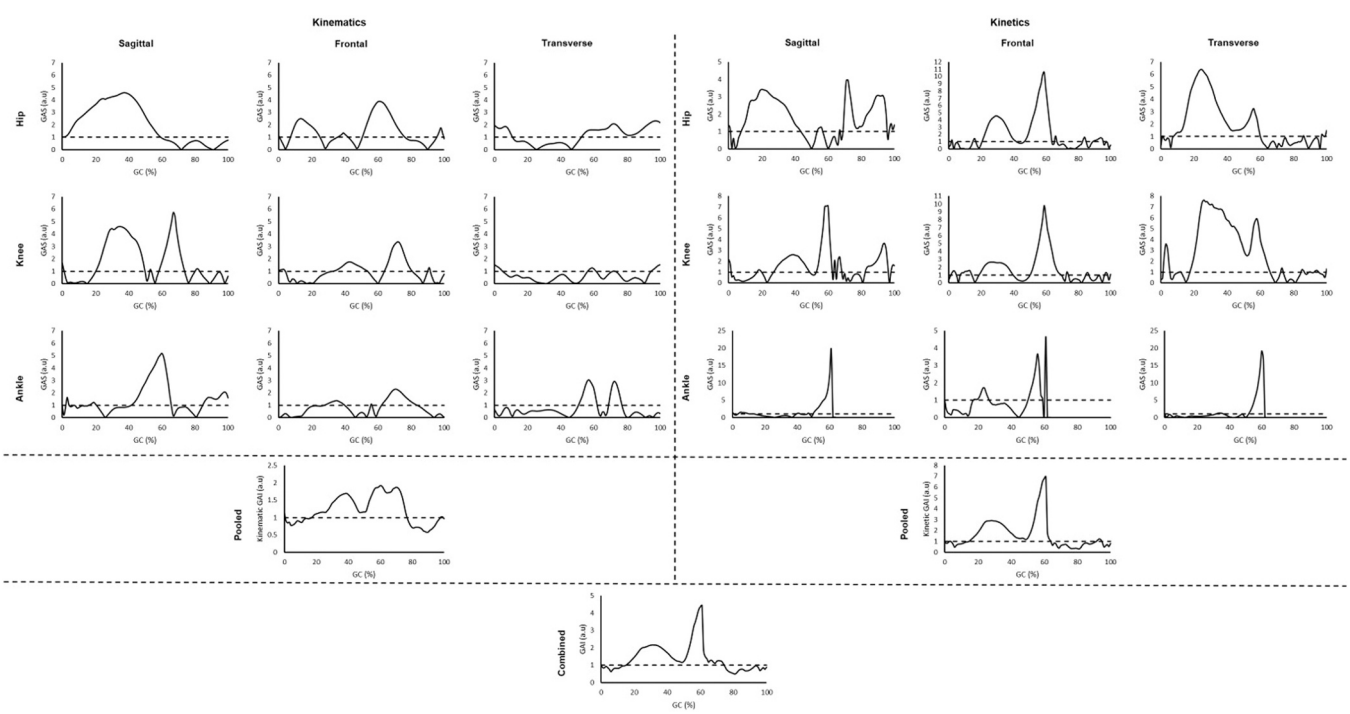

Fig. 1. Representative Gait Abnormality Scores (GAS) for hip, knee and ankle joint kinematics and kinetics, in the sagittal, frontal and transverse planes over the walking gait cycle (GC), with the resulting kinematic, kinetic and combined Gait Abnormality Index's (GAI) presented for an individual knee osteoarthritis patient (59 year old male, 1.65 m and 83 kg). Values below the horizontal dashed line fall within normative ranges.

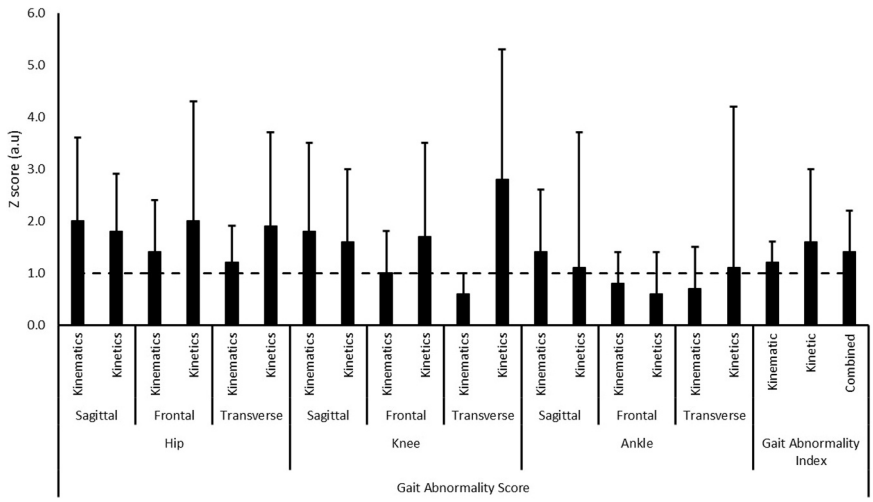

Fig. 2. Modified Movement Analysis Profile (MAP) developed from the concept proposed by Baker et al., (2009). Average and standard deviations associated with each Gait Abnormality Score (GAS) for hip, knee and ankle kinematics and kinetics, with the resulting kinematic, kinetic and combined Gait Abnormality Index (GAI) displayed for an individual knee osteoarthritis patient (59 year old male, 1.65 m and 83 kg). Values below the horizontal dashed line fall within normative ranges.

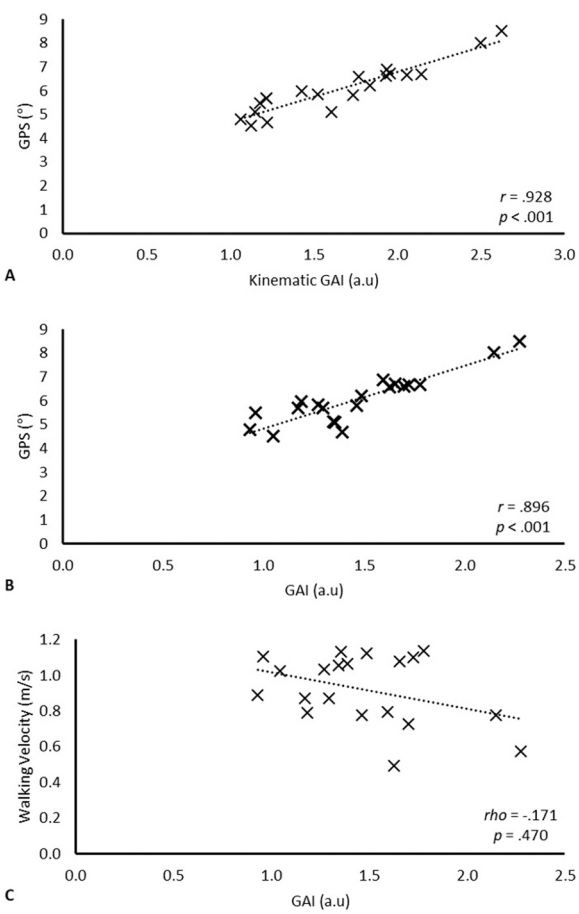

Fig. 3. Correlations between the kinematic components only of the Gait Abnormality Index (GAI) and the Gait Profile Score (GPS) (A), the combined GAI and GPS (B), and combined GAI and walking velocity (C).

include kinematic and kinetic data within the final GPS metric. The GAI proposed within this study reincorporates a standardisation element within the calculation, via conversion to z-scores, in a similar manner to the GDI. This standardisation removes the units of measurement, with the data effectively reported as standard deviations away from the healthy control group mean, enabling the simultaneous inclusion of both kinematic and kinetic data. In contrast, to the GDI, absolute z-scores are calculated for the GAI removing the apparent assumption that scores above the healthy control group mean will denote normal or healthy patterns, as opposed to alternate pathological or abnormal motion or loading. With both the GAI and GPS underpinned by the same distance-based calculation the strong positive correlations between both the kinematic only and combined variations of the GAI and the GPS are unsurprising (Fig. 3A & B). As such, and in accordance with the argument proposed by Baker et al., [4], any evidence validating the GPS would validate the GAI and vice versa.

The GAI appears to provide a reliable means of summarising gait kinematics and kinetics. Good test-retest reliability was reported for the GAI, with the associated SEM 0.16 a.u. It is likely that the reliability of the GAI, and in fact all summary gait metrics, is inherently linked to the reliability of the joint kinematic and kinetic data which act as the input for the metrics calculation. As such the reliability estimated for the GAI may be limited to the total hip arthroplasty population assessed and also to the specific six degrees-of-freedom model utilised within this study.

The GAI values for both the knee osteoarthritis and total hip arthroplasty groups are above 1, demonstrating that both groups display abnormal gait kinematics (Fig. 4). These findings are in agreement with the previous literature, with significant differences in lower limb kinematics and kinetics reported for knee osteoarthritis [15,16] and total hip arthroplasty [17–19] groups compared to healthy controls during walking. While, Kosbar et al [7] and Jensen et al., [20] have reported that gait patterns of knee osteoarthritis and total hip arthroplasty patients, respectively, are roughly 1 – 2 standard deviations away from normal, based upon GDI values. Deviations from normal of these magnitudes are comparable to those reported by both patient groups within this study, based upon the GAI. Speculatively, the larger GAI reported by the knee osteoarthritis patients may be due to post-operative enhancements in gait mechanics within the total hip arthroplasty patients, with previous studies [18,19] reporting improved lower limb kinematics and kinetics post-operatively. While the significant difference in GAI values between the knee osteoarthritis and total hip arthroplasty groups suggests that the metric is sensitive to different populations and pathologies.

That the GAI is independent of walking velocity (Fig. 3C) is beneficial due to velocity often differing between healthy and pathological populations [16–20]. Differences in walking velocity between healthy and pathological populations have been identified as a major challenge within clinical gait analysis, due to significant associations between joint

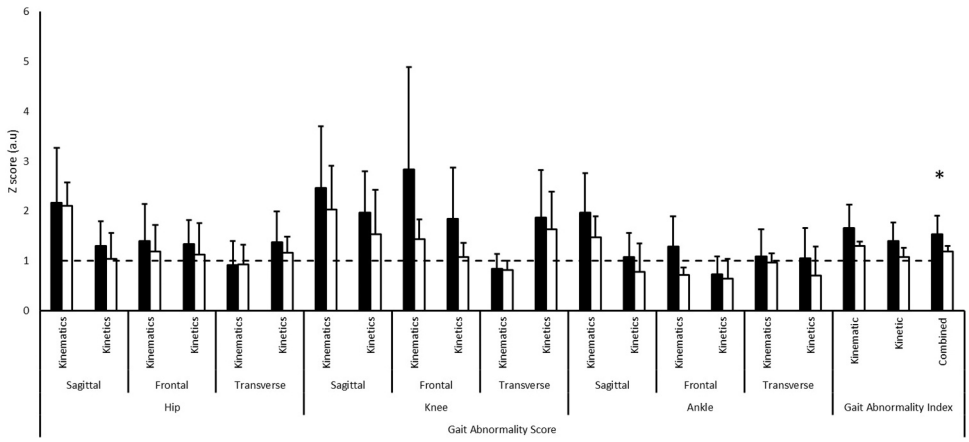

Fig. 4. Comparison of Gait Abnormality Scores and the Gait Abnormality Index between knee osteoarthritis (n = 20, black) and total hip arthroplasty (n = 10, white) patients. Values below the horizontal dashed line fall within normative ranges and * denotes $p < .05$.

angles and moments, and velocity [21]. As such the GAI may provide a means of overcoming the difficulties in identifying velocity and pathology related alterations in gait mechanics. While walking velocity is not significantly related to GAI scores, differences in stance duration do appear to be influential in the calculation of this metric based upon visual exploration of the GAS and GAI scores over the gait cycle. For example, the longer stance durations displayed by the individual knee osteoarthritis patient and this group (Table 1) result in larger GAS for joint moments, especially at the knee and ankle, around toe off (Fig. 1; Sup Figure 6). Furthermore, these larger kinetic alterations contribute to large changes in GAI around this time period. However, as this increase in stance duration is likely due to the knee osteoarthritis patients looking to reduce time spent in single limb support to enhance their dynamic stability it is of functional relevance [22]. The suggestion that stance duration is influential in the calculation of the GAI is further supported by the smaller magnitude GAI and GAS around toe off for the total hip arthroplasty group (Supplementary Figure 6), who did not display increased stance duration compared to the healthy controls (Table 1).

5. Conclusion

The GAI offers a simple, easy to calculate summary metric for 3DGA, which can simultaneously include kinematic and kinetic data. The GAI provides an evolution of the MAP concept developed previously [4], enabling the contribution of relevant joint kinematics and kinetics to the overall score to be explored. The GAI demonstrated good validity, good reliability and a small SEM, and was sensitive to different pathological conditions.

Declaration of Competing Interest

The authors report no conflict of interests related to this study.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.gaitpost.2023.07.281.

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